

AGRIAI 2.0

INTELLIGENT FARMING ASSISTANT FOR GHANA

Volume IV. Code Path Atlas

Which files execute when a farmer or admin clicks

Prepared and authored solely by Thaddeus Nii Teiko Tagoe

Computer Science, University of Ghana

Founder, designer, and sole engineer of AgriAI

Repository: ThaddeusHackz/agriai-v2

Document date: 18 August 2026

Classification: Internal forensic architecture pack

This document explains how AgriAI was built, which source files make the AI work, and how the public and admin APIs stay reliable when a provider is missing. Every claim below is taken from the live codebase, not from marketing copy.

1. How to read this atlas

Each path is a real call chain from a UI file to a library. Use it as a study map.

2. Path A. Typed question in the chat box

- components/Chat.tsx collects message, language, mode, webSearch, history, sessionId.
- fetch POST /api/chat with JSON.
- app/api/chat/route.ts validates and calls trackQuestion + trackMessage.
- Optional: lib/search.ts searchWeb then contextBlock.
- getDB().settings.chat selects model, temperature, maxTokens, and the mode prompt.
- If geminiConfigured: lib/ai.ts geminiGenerateStream, SSE delta events.
- Else or on failure: lib/cloudflare.ts cloudflareChat.
- Else: lib/ai.ts localAnswer using lib/db.ts knowledge.
- mutate() writes both turns. SSE done. Chat.tsx renders markdown and sources.

3. Path B. Photo of a sick leaf

- components/DiseaseDetector.tsx reads the file as a data URL.
- POST /api/vision { image, language }.
- app/api/vision/route.ts size and language checks.
- lib/ai.ts detectCropDisease: Gemini multimodal JSON, or FALLBACK_DISEASE.
- UI shows detected name, confidence, description, treatment steps, demo badge.

4. Path C. Crop Visualizer

- components/CropStudio.tsx POST /api/generate { prompt }.
- SAFE_SUBJECTS gate in app/api/generate/route.ts.
- lib/cloudflare.ts cloudflareImage Flux, else lib/ai.ts geminiImage.
- JSON { ok, image, provider }. Farmer downloads the PNG.

5. Path D. Microphone

- Chat.tsx prefers window SpeechRecognition.
- Fallback: MediaRecorder blob to POST /api/transcribe multipart audio.
- Gemini 2.5 Flash inlineData audio, returns { text } into the input box.

6. Path E. Speaker

- Chat.tsx POST /api/tts { text, language, speed }.
- ElevenLabs voice map, audio/mpeg response, Audio element plays it.

7. Path F. Weather strip

- components/WeatherSection.tsx GET /api/weather.
- lib/openweather.ts then Open-Meteo then curated.
- Advice strings stay Ghana season aware even in fallback.

8. Path G. Admin changes the AI personality

- components/admin/AITab.tsx edits model, temperature, prompts.
- components/admin/api.ts PUT /api/admin/settings.
- lib/auth.ts accepts the cookie. mutate() writes settings.chat.
- Next farmer message uses the new prompt with no rebuild.

9. Path H. First boot

- instrumentation.ts runs hydrateFromPostgres if data/db.json is absent.
- lib/db.ts load() or defaultDB() seeds prices, knowledge, bcrypt admin.

- app/layout.tsx and lib/site-context.tsx feed public config to components.

10. Source file roster for AI enablement

File	Why AI cannot work without it
lib/ai.ts	Gemini client, stream, vision, image, localAnswer.
lib/cloudflare.ts	Fallback LLM and Flux.
lib/search.ts	Citations and grounding text.
lib/env.ts	Secret resolution and aliases.
lib/languages.ts	Six language system addenda.
lib/db.ts	Prompts, knowledge, persistence.
app/api/chat/route.ts	Orchestrates the ladder and SSE.
app/api/vision/route.ts	Disease HTTP edge.
app/api/generate/route.ts	Studio HTTP edge.
app/api/transcribe/route.ts	Voice to text HTTP edge.
app/api/tts/route.ts	Text to voice HTTP edge.
components/Chat.tsx	The farmer facing brain.

11. Closing statement of authorship

Volumes I to IV map construction, intelligence, APIs, and call chains. Volumes V and VI add the security threat model and the sequence posters. The whole pack is authored by Thaddeus Nii Teiko Tagoe, sole creator of AgriAI and of this documentation.